



Physics Worksheet

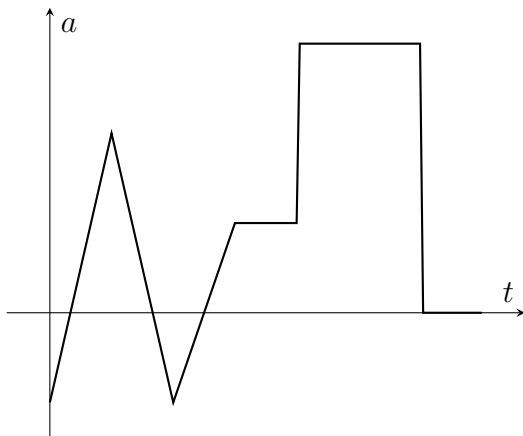
Kinematics

Name: _____ Student #: _____ Student Email #: _____

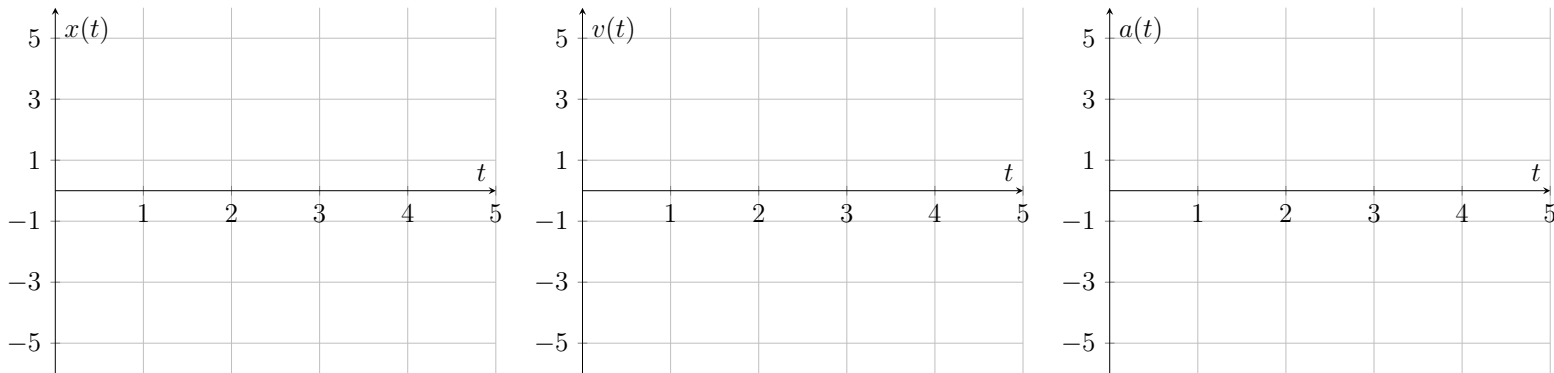
1. The acceleration of a cricket jumping sideways is observed to increase linearly with time, that is, $ax(t) = a_0 + jt$, where a_0 and j are constants. What can you say about the velocity of the cricket as a function of time?

- (a) It is constant
- (b) It increases linearly with time ($v(t) \propto t$)
- (c) It increases quadratically with time ($v(t) \propto t^2$)
- (d) It increases with the cube of time ($v(t) \propto t^3$)

2. Plot below the $v(t)$ and $x(t)$ for the given $a(t)$ plot. Assume $v_0 = 0$ and $x_0 = 0$



3. Below, plot the function $x(t) = 2x^{\frac{1}{2}}$, plot this as well as $v(t)$ and $a(t)$ of the same function



4. A large motor is running at 4000 rpm with an internal linear velocity of $v = 20 \text{ m s}^{-1}$. What is the radius R of the motor and the frequency f at which the motor is rotating?
5. Enzo is driving a remote control car on a train that has just begun to move after making a stop at a station
- (a) If the train is accelerating at $a = 2 \text{ m s}^{-2}$ at $t = 0 \text{ s}$ and Enzo's toy car is going at a constant speed of $v = -3 \text{ m s}^{-1}$, what is the toy car's position as a function of time as seen from the stationary platform of the train station?
- (b) At what point in time will the toy car appear to be moving in the same direction as the train?